

Lab3

Analyze the requirement (using Template 1), then write the test cases (using Template 2) for **Loan Eligibility Calculator** function

Screen: Loan Eligibility Calculator

Input Fields

Field	Type	Description
Age	Input number field	The system will use this value to determine whether the applicant meets the age requirement for loan eligibility
Monthly Income	Input number field	User enters their total monthly income in USD , This value is used to calculate the Debt Ratio
Existing Debt	Input number field	User enters their current total monthly debt payment in USD. This value is used to calculate the Debt Ratio
Unemployed	Checkbox	Indicates whether the user is currently unemployed. Default value: Unchecked

Buttons

Name	Description
Clear	When the user clicks Clear : • Age field is cleared • Monthly Income field is cleared • Existing Debt field is cleared • Unemployed checkbox is unchecked
Check Eligibility	When the user clicks Check Eligibility : 1. System performs input validation. 2. If any input is invalid → show the corresponding validation message below the corresponding field. 3. If all inputs are valid → system calculates Debt Ratio. 4. After calculation, the system displays:

	Loan Eligibility: <Result> Debt Ratio: <Value> Example Output: Loan Eligibility: Medium Risk Debt Ratio: 0.35
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Function Detail

1. Input Validation

Age

- Required field
- Must be a **number between 18 and 65**

Validation messages:

- If invalid, the system will show a red message below the Age field:
“Please enter an age between 18 and 65.”

Monthly Income

- Required field
- Must be a **number greater than 0**

Validation messages:

- If invalid, the system will show red message below the Monthly Income field :
“Monthly Income must be greater than 0.”

Existing Debt

- Required field
- Must be a **number greater than or equal to 0**

Validation messages :

- If invalid, the system will show red message below the Existing Debt field:
“Existing Debt must be greater than or equal to 0.”

2. Business rules

2.1 Loan Eligibility Formula

Debt Ratio is calculated as:

Debt Ratio = Existing Debt / Monthly Income

2.2 Eligibility Rules

a. Special Rule

If **Unemployed checkbox = Checked**, loan eligibility result will always be
(Regardless of the Debt Ratio): Not Eligible

b. Normal Eligibility Rules

If **Unemployed checkbox = Unchecked**, loan eligibility result is calculated based on the following rules:

Debt Ratio	Result
> 0.6	Not Eligible
0.4 – 0.6	High Risk
0.2 – 0.4	Medium Risk
< 0.2	Low Risk